

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – VII
PAPER –1
TEST DATE: 07-08-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 – 04, 20 – 23, 39 – 42): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
Section – A (05 – 10, 24 – 29, 43 – 48): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

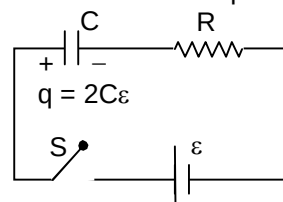
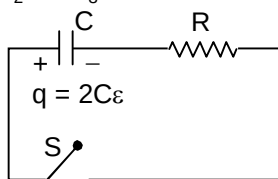
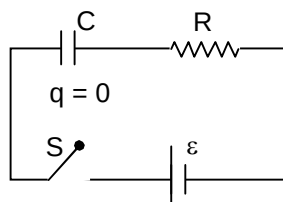
Physics

PART – I

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

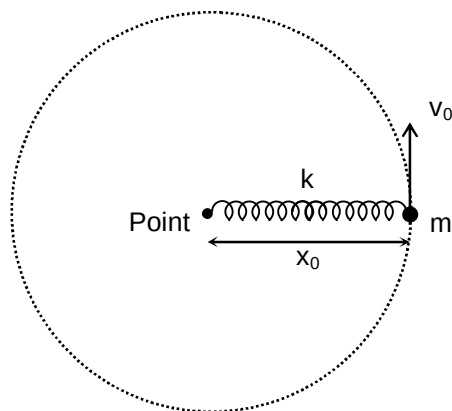
1. Three circuits are shown in their initial conditions. In every circuit switch is closed at $t = 0$. Then currents at time t are observed i_1 , i_2 and i_3 in clockwise direction. Choose the correct option.



- (A) $i_1 + i_2 = i_3$
 (B) $i_1 + i_2 > i_3$
 (C) $i_1 + i_2 < i_3$
 (D) $i_1 + i_3 = i_2$
2. A stationary source emits sound of frequency $f_0 = 492$ Hz. The sound is *reflected* by a large car approaching the source with a speed of 2 ms^{-1} . The reflected signal is received by the source and superposed with the original. The beat frequency of the resulting signal is (Given that the speed of sound in air is 330 ms^{-1} and the car reflects the sound at the frequency it has received).
- (A) 2 Hz
 (B) 4 Hz
 (C) 6 Hz
 (D) 8 Hz
3. A uniform rod is kept fixed at one end at rigid wall and placed on smooth horizontal floor. It is pulled by a force under proportionality limit horizontally and released. Assume there is no energy loss and rod performs small oscillations. Then time period of oscillations of rod will be (Take mass m , length ℓ , Young's modulus Y area of cross section).

- (A) $2\pi\sqrt{\frac{m\ell}{3AY}}$
 (B) $2\pi\sqrt{\frac{m\ell}{AY}}$
 (C) $2\pi\sqrt{\frac{m\ell}{2AY}}$
 (D) $2\pi\sqrt{\frac{4m\ell}{3AY}}$

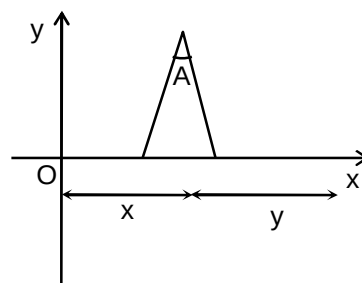
4. A hypothetical massless spring having negligible relaxed length and follows the relation between force F and extension x as $F = cx^{-2}$, where c is constant and x is extension in spring. Spring can only be broken if it stretched upto a very large length. A point mass is performing circular motion with help of spring as shown, with speed v_0 . The velocity needed for mass to break the spring is close to:
- (A) $2v_0$
 (B) $\sqrt{2}v_0$
 (C) $3v_0$
 (D) $2\sqrt{2}v_0$



Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

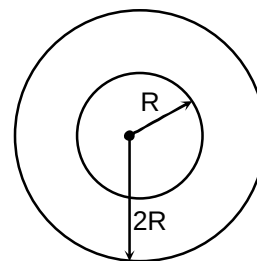
5. A thin prism having prism angle A is placed on x -axis as shown in the figure having variable refractive index along y -direction as $\mu = \frac{h}{k} + 1$, where k is positive constant and h is height from base. A point object is placed at origin and its image forms at y distance on another side then choose the correct combination(s).



- (A) $x = \frac{2k}{A}$; $y = \frac{2k}{A}$
 (B) $x = \frac{3k}{A}$; $y = \frac{3k}{2A}$
 (C) $x = \frac{3k}{A}$; $y = \frac{k}{2A}$
 (D) $x = \frac{k}{2A}$; $y = \frac{3k}{2A}$
6. Two identical bubbles of radius r and surface tension S are isolated in vacuum. They come in contact and stick together making a common surface. The radius of common surface is r and loss in surface energy is ΔU then
- (A) $r = \frac{r_0}{2}$
 (B) $r = \infty$
 (C) $\Delta U = \frac{5}{2} S\pi r^2$
 (D) $\Delta U = \frac{3}{2} S\pi r^2$

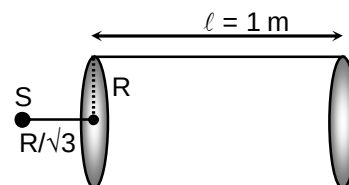
7. Two bubbles contain same ideal gas such that one is trapped inside other. System is kept in vacuum. Ratio of densities of inner and middle gas is x and the ratio of number of moles is y then value of $\frac{x}{y}$ is (Assume temperature of both regions are same)

- (A) 3
(B) 7
(C) 9
(D) 14



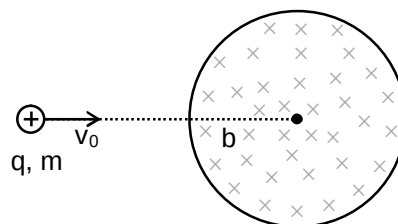
8. A conducting cylinder has curved surface covered with insulated layer. Flat surfaces are perfect absorbers. Flat surface area is $\frac{3}{17} \text{ m}^2$. A point light source placed at some distance from left face. Temperature of surrounding is 0 K. In steady state temperature of left face is 200 K and right face 100 K. Then (Stefan's constant = $\frac{17}{3} \times 10^{-8} \text{ W/m}^2 \text{ K}^4$)

- (A) Power emitted by source is 68 W
(B) Power emitted by source is 18 W
(C) Thermal conductivity of cylinder is 0.057 W/mK
(D) Thermal conductivity of cylinder is 0.023 W/mK



9. A charge particle is projected towards centre of a circular region of uniform magnetic field having radius b . The deviation of charge varies with v_0 as

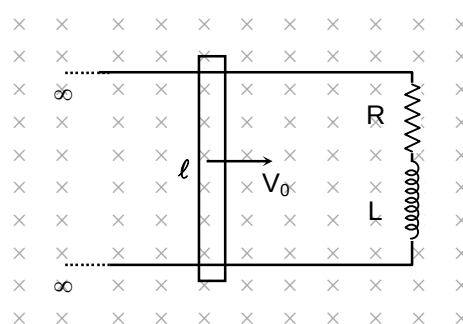
- (A) linearly for small v_0
(B) inversely for large v_0
(C) linearly for large v_0
(D) inversely for small v_0



10. A conducting rod can slide on smooth conducting rails connected with series inductor and resistor at the end. Rod is given initial velocity v_0 . Initially there is no current in circuit when rod has covered forward displacement s and current in resistor is i , speed of rod is found as :

$v = v_0 - \alpha s + \beta i$
value of α and β are:

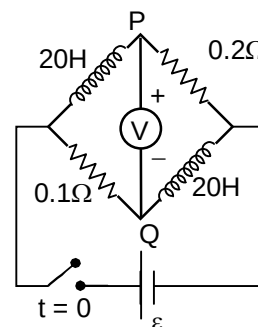
- (A) $\alpha = \frac{B^2 \ell^2}{mR}$
(B) $\alpha = \frac{B^2 \ell^2}{2mR}$
(C) $\beta = \frac{B\ell L}{3mR}$
(D) $\beta = \frac{B\ell L}{mR}$



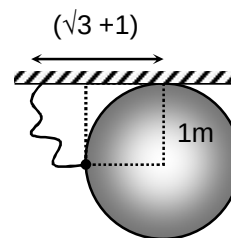
Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

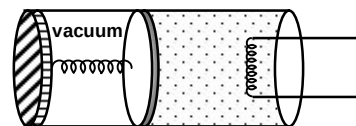
11. In the circuit shown switch is initially open for long time and closed at $t = 0$. Time (in sec) when voltmeter shows zero reading will be? [Take $\ln(0.62) = -0.49$ and $\sqrt{5} = 2.236$]



12. A disc having radius 1m is held in contact with ceiling. A light rope is tied at circumference of disc and kept the point at same level as centre the other end of rope is tied at ceiling at $(\sqrt{3} + 1)$ m distance from contact point disc is released from rest. The rope length is $\sqrt{7}$ m. The rope snaps due to impulse and disc momentarily moves horizontally, just after snapping. The angular velocity of disc after snapping is $k\sqrt{20}$ rad/s. Find the value of k.



13. Monoatomic gas is kept in equilibrium in an insulated cylinder of length L. Piston is attached with an ideal spring as shown which has relaxed length L. The spring is fixed at one end of cylinder. The gas is supplied very small amount of heat dQ . Piston gets very small displacement in this process. The work done by gas is $\frac{dQ}{N}$. The value of N is



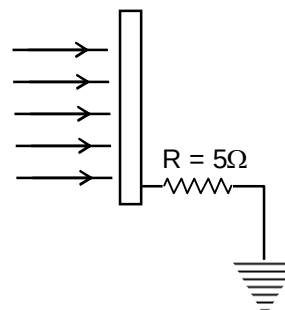
Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15

Question Stem

A large metal plate having area 1 m^2 is exposed to light of intensity 6.62 W/m^2 . Frequency of light is $3.2 \times 10^{15} \text{ Hz}$ which is greater than threshold value of photoemission. Emitted electrons are swept away immediately after emission. ($h = 6.62 \times 10^{-34}$)



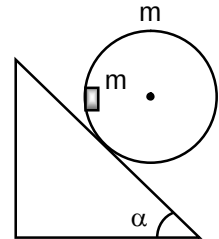
14. The current through resistor in steady state in ampere is.....

15. The potential of plate in steady state in volt is

Question Stem for Question Nos. 16 and 17

Question Stem

The inclined plane with angle α has a cylinder resting on it. A small block of same mass is attached inside it. Assume incline plane is highly rough. The system is in equilibrium.

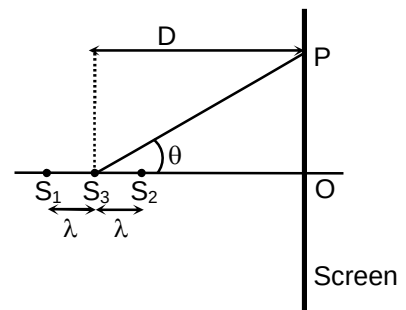


16. What is the maximum value of α so that system can be in equilibrium (in radian)
17. Minimum friction coefficient of inclined plane to achieve previous condition.

Question Stem for Question Nos. 18 and 19

Question Stem

Two identical sources of light are placed at λ separation from third identical source collinearly as shown in the figure. Their light superimposed and screen placed at very large distance. Wavelength of light emitted by each source is λ .



18. If intensity at centre 'O' is I_0 and intensity at $\theta = 60^\circ$ is αI_0 . Then find value of α .
19. If wavelength of each source becomes 8 times then intensity at centre 'O' is N times due to the intensity of each source. Find N.

Chemistry

PART – II

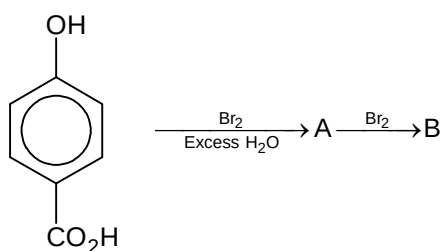
Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

20. de-Broglie wavelength of an electron in the first Bohr orbit of H-atom is

- (A) $1.7 \text{ \AA}$
 (B) $3.3 \text{ \AA}$
 (C) $6.8 \text{ \AA}$
 (D) $5.5 \text{ \AA}$

21.



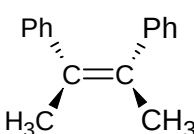
The structure of 'B' would be?

- (A)
- (B)
- (C)
- (D)

22. What is the % decrease in vapour pressure (relative to solvent) when saturated aqueous solution of silver phosphate is prepared? (Assume density = 1 gm/ml) [Given K_{sp} of silver phosphate = 2.7×10^{-15}]
- (A) 9×10^{-4}
 (B) 1.6×10^{-4}
 (C) 7.2×10^{-4}
 (D) 2.44×10^{-4}
23. For the reaction $A(g) \rightleftharpoons 2B(g)$, the equilibrium partial pressures of gases A and B are 4 atm. and 2 atm. respectively. If equilibrium mixture is slowly and isothermally compressed to 12 atm, maintaining equilibrium, then the new equilibrium pressure (in atm.) of B (g) is
- (A) 3 atm
 (B) 4 atm
 (C) 6 atm
 (D) 8 atm

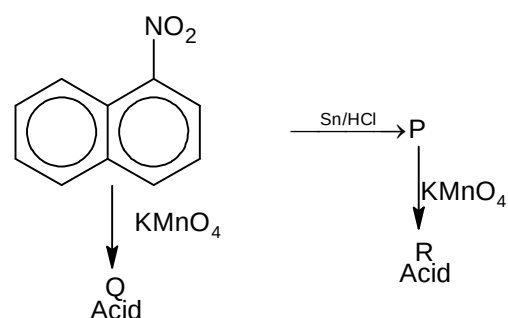
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

24.  $\xrightarrow{x}$ Meso – 2,3 – diphenylbutane

'x' can be

- (A) $NH_2 - NH_2 + H_2O_2$
 (B) (i) B_2H_6 in THF; (ii) CH_3CO_2H
 (C) H_2, Pd
 (D) (i) B_2H_6 in THF; (ii) H_2O_2 / OH^-

25. 

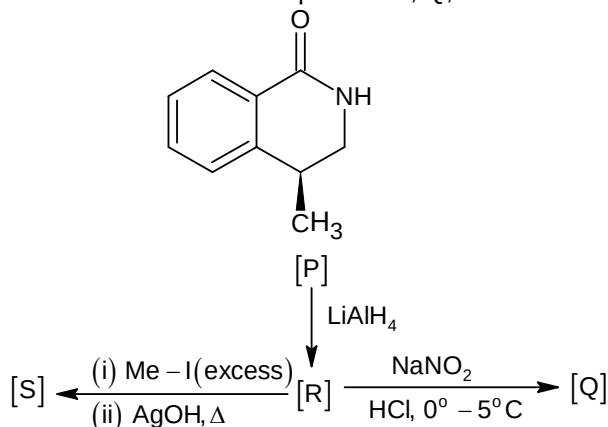
Select the correct statements related to 'Q' and 'R'

- (A) Both Q and R contain Nitrogen atom
 (B) Nitrogen atom is present in Q but absent in R
 (C) Both Q and R are dibasic acid
 (D) Nitrogen atom is absent in both Q and R
26. Which of the following statements is/are correct?
- (A) Bakelite is a polyamide
 (B) All biodegradable polymers are natural polymers
 (C) Teflon is a thermoplastic polymer
 (D) Nylons are condensation polymers or copolymers

27. Which of the following statements is/are correct?
- (A) Cationic part of liquid Cl_2O_6 is planar
 (B) Highest occupied MO of ClO_2 is an incompletely filled antibonding orbital
 (C) HNO_2 and HClO_3 both disproportionate on heating
 (D) Sodium perchlorate is better bleaching agent than sodium hypochlorite
28. Which of the following statement/s is/are correct?
- (A) For a nuclide ${}^A_Z\text{X}$, A does not change but Z decreases by one unit in K-electron capture
 (B) In the following decay chain, A and B are isodiaphers, A and D are isotopes

$$A \xrightarrow{-\alpha} B \xrightarrow{-\beta} C \xrightarrow{-\beta} D$$

 (C) ${}_{88}\text{Ra}^{226} \xrightarrow{-\alpha} A \xrightarrow{-\alpha} B$; B belongs to period 6, group 14
 (D) Radio active disintegration rate is not affected by the number of atoms of the radio active element undergoing disintegration
29. With reference to the scheme given below, which of the following statement(s) is/are INCORRECT about compounds P, Q, R and S?

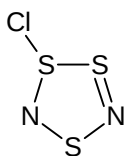


- (A) [P] release ammonia when boiled with concentrate NaOH solution
 (B) [Q] is a yellow oily liquid and is chiral
 (C) [S] shows geometrical isomerism
 (D) [S] undergoes ring nitrosation on treatment with NaNO_2 and dil. HCl

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

30. Consider the following structure for the ion $\text{S}_3\text{N}_2\text{Cl}^+$ found in the salt $(\text{S}_3\text{N}_2\text{Cl}^+)\text{Cl}^-$,



How many atoms in the above picture show non-zero formal charges (place non bonding electrons as necessary)

31. 'x' g of monobasic acid HX. When dissolved in 100 g of water, lower freezing point by 0.279°C . If 0.2 g of the same acid is (HX) completely neutralized by 20 ml of N/10 NaOH solution and if $\frac{\lambda_{\text{HX}}(\text{aq. solution})}{\Lambda_{\text{HX}}^{\circ}} = 5 \times 10^{-1}$; then find the name of 'x': (k_f of $\text{H}_2\text{O} = 1.86 \text{ Kkg mol}^{-1}$)
32.
$$\text{H} - \overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{H} \xrightarrow{\text{Conc. NaOH}} \text{A} + \text{B}$$

If the equivalent weight of HCHO in the above reaction is E, then the value of E/5 is

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 33 and 34

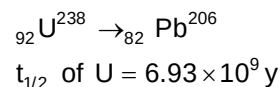
Question Stem



33. Number of possible isomers of A which can produce diastereomeric products is 'x' then value of $\frac{2x}{5}$ is _____
34. If number of possible structures of A which can give positive iodoform test is 'a' and number of possible structural isomers which give positive Fehlings Test is b then value of $\frac{a}{b}$ is _____

Question Stem for Question Nos. 35 and 36

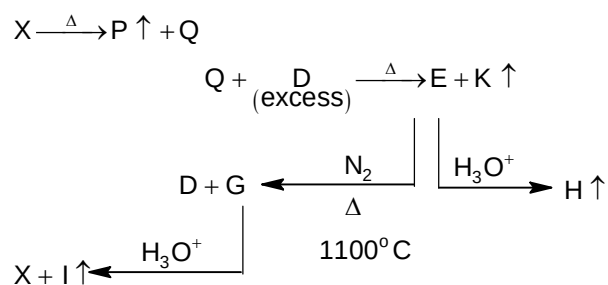
Question Stem



35. An uranium mineral contains 20.6 of lead for each 100 g of uranium present. If age of mineral is $x \times 10^9$ years, then determine x. [Assume no lead is present initially; $\log 1.238 = 0.11$]
36. Find the mass of lead in g from 1 kg of U^{238} , during the age of the mineral.

Question Stem for Question Nos. 37 and 38

Question Stem



Gas P makes lime water milky. K is a poisonous gas. H is used in welding.

37. Molar mass ratio of 'G' and that of 'x' would be _____
38. Molar mass ratio of 'H' and that of I would be _____

Mathematics

PART – III

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

39. For a complex number $z = \cos 20^\circ + i \sin 20^\circ$, the value of $|z + 2z^2 + 3z^3 + \dots + 18z^{18}|^{-1}$ is

- (A) $\frac{\cos 80^\circ}{9}$
- (B) $\frac{\cos 80^\circ}{18}$
- (C) $9 \operatorname{cosec} 10^\circ$
- (D) $\frac{\cos 70^\circ}{9}$

40. $\sum_{k=1}^{\infty} \cot^{-1} \left(\frac{k^4 + 5}{4k} \right)$ is equal to

- (A) $\tan^{-1} \left(\frac{3}{4} \right)$
- (B) $\frac{\pi}{4}$
- (C) $\frac{\pi}{4} + \cot^{-1} 2$
- (D) $\frac{\pi}{4} + \tan^{-1} 2$

41. The solution of the differential equation $(x - 1)dy + y dx = x(x - 1)y^{1/3} dx$ is given by $y^{2/3} = k(x - 1)^2 + m(x - 1) + c(x - 1)^{-2/3}$, then the value $k + m$ is

- (A) $\frac{39}{40}$
- (B) $-\frac{3}{20}$
- (C) $\frac{13}{2}$
- (D) $\frac{13}{20}$

42. If a polynomial function $f(x)$ satisfies $f(f(f(x))) = 125x + 31$, then $f(0)$ is

- (A) 0
- (B) -1
- (C) 1
- (D) 5

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

43. Circles are drawn on the chords of rectangular hyperbola $xy = 16$ parallel to the line $y = x$ as diameters. All such circles pass through two fixed points whose coordinates are
 (A) $(4\sqrt{2}, 4\sqrt{2})$
 (B) $(4, 4)$
 (C) $(-4, -4)$
 (D) $(-4, 4)$
44. Let $S_n = \frac{2}{4!} + \frac{7}{5!} + \frac{14}{6!} + \frac{23}{7!} + \dots$ upto n terms. If t_n denote the n th term, then which of the following is/are correct?
 (A) $T_{10} = \frac{119}{13!}$
 (B) $T_{14} = \frac{225}{17!}$
 (C) $S_{20} = \frac{1}{6} - \frac{21}{23!}$
 (D) $\lim_{n \rightarrow \infty} S_n = \frac{1}{3}$
45. Given $I_1 = \int_0^{\pi/2} \cos(\sin x) dx$, $I_2 = \int_0^{\pi/2} \sin(\cos x) dx$, $I_3 = \int_0^{\pi/2} \cos x dx$, then which of the following is/are INCORRECT?
 (A) $I_1 > I_3 > I_2$
 (B) $I_1 > I_2 > I_3$
 (C) $I_3 > I_1 > I_2$
 (D) $I_3 > I_2 > I_1$
46. Let $f(x) = \lim_{n \rightarrow \infty} \frac{kx^{2n} \log_e(4 - |x|) + m \log_e(2 + |x|)}{m - kx^{2n+1}}$ where $m, k \in \mathbb{R}$ and $mk \neq 0$. Then which of the following is/are CORRECT?
 (A) $\lim_{x \rightarrow 1} f(x)$ exists
 (B) $\lim_{x \rightarrow -1} f(x)$ exists
 (C) $\lim_{x \rightarrow 1} f(x)$ does not exist
 (D) $\lim_{x \rightarrow -1} f(x)$ does not exist
47. A ray of light comes along line L and strikes the plane mirror kept along the plane $P = 0$ at point B . $A(2, 1, \lambda)$ is a point on the line L whose image about $P = 0$ is A' . It is given that line L is $\frac{x-2}{3} = \frac{y-1}{4} = \frac{z-6}{5}$ and plane $P = 0$ is $x + y - 2z = 3$, then
 (A) the coordinates of A' are $(6, 5, -2)$
 (B) the coordinates of A' are $(6, 5, 2)$
 (C) equation of the reflected ray is $\frac{x+10}{4} = \frac{y+15}{5} = \frac{z+14}{3}$
 (D) equation of the reflected ray is $\frac{x+10}{4} = \frac{y-15}{4} = \frac{z+2}{3}$

48. From a point P representing complex number z_1 on the curve $|z| = 2$, two tangents are drawn from P to the curve $|z| = 1$ meets at $A(z_2)$ and $B(z_3)$, then
- (A) centroid of $\triangle ABP$ will be on the curve $|z| = 1$
- (B) $\text{Arg}\left(\frac{z_2}{z_3}\right) = \pm \frac{2\pi}{3}$
- (C) $\left(\frac{4}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right)\left(\frac{4}{\bar{z}_1} + \frac{1}{\bar{z}_2} + \frac{1}{\bar{z}_3}\right) = 9$
- (D) orthocentre and circumcentre of $\triangle ABP$ will coincide

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.*

49. A differentiable function f satisfies the relation $f(x+y) = f(x) + f(y) + 2xy(x+y) - \frac{1}{2} \forall x, y \in \mathbb{R}$ and $\lim_{h \rightarrow 0} \left(\frac{2f(h) - 1}{4h} \right) = 1$, then the value of $[f(1)]$ is (where $[.]$ denotes greatest integer function)
50. $(1 + ax + bx^2)^4 = a_0 + a_1x + a_2x^2 + \dots + a_8x^8$ where $a, b, a_0, a_1, a_2, \dots, a_8 \in \mathbb{R}$ and $a_0 + a_1 + a_2 \neq 0$ and $\begin{vmatrix} a_0 & a_1 & a_2 \\ a_1 & a_2 & a_0 \\ a_2 & a_0 & a_1 \end{vmatrix} = 0$, then the value $\frac{5a}{b}$ is
51. The tangent and normal to the parabola $y^2 = 8x$ at $(2, 4)$ intersect the line $y = 3 - kx$ at points P and Q respectively. If PQ subtend a right angle at the vertex of the parabola, then absolute value of sum of all possible values of k is

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 52 and 53

Question Stem

Given $p, q \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$. Consider the system of equations

$$x + y + z = 4; 2x + y + 3z = 6; x + 2y + pz = q$$

Let A : denote number of ordered pairs (p, q) so that the system of equation has unique solution
 B : denote number of ordered pairs (p, q) so that system of equations has no solution
 C : denote number of ordered pairs (p, q) so that system of equations has infinite solution

52. The value of A is
53. The value of $A + B + C$ is

Question Stem for Question Nos. 54 and 55

Question Stem

Let $f(x)$ be defined as $f(x) = \begin{vmatrix} \sin(x+a) & \sin(x+b) & \sin(x+c) \\ \cos(x+a) & \cos(x+b) & \cos(x+c) \\ \cos(b-a) & \cos(c-b) & \cos(a-c) \end{vmatrix}$ and let matrix $P = \begin{bmatrix} \cos\left(\frac{\pi}{9}\right) & \sin\left(\frac{\pi}{9}\right) \\ -\sin\left(\frac{\pi}{9}\right) & \cos\left(\frac{\pi}{9}\right) \end{bmatrix}$

where a, b, c be non-zero real numbers such that $(aP^6 + bP^3 + cI)$ is a zero matrix and where I is identity matrix of order 2. If $f(3) = \lambda \neq 0$

54. The value of $\frac{\sum_{k=1}^3 f(k)}{f(3)}$ is

55. The absolute value of $\frac{2b}{c}$ is

Question Stem for Question Nos. 56 and 57

Question Stem

Consider $f(x) = \left| \sin\left((2p_1 - 1)\frac{\pi}{6}\right)x \right| + \left| \cos\left(\frac{p_2\pi}{6}\right)x \right|$. A bag contains tickets numbered $\{1, 2, 3, 4, 5, 6, 7, 8\}$.

Two tickets are drawn one by one with replacement. Let p_1 and p_2 be numbers obtained on first and 2nd draw respectively

56. The probability that $f(1)$ is integer is

57. The probability that $f(2)$ is irrational